



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-1.2

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Subject Name: DAA

Subject Code: 23CSH-301

1. **Aim:** Code implement power function in $O(\log n)$ time complexity.
2. **Objective:** To implement power function in $O(\log n)$ time complexity.
3. **Input/Apparatus Used:** In this program , power is divided by 2 in order to get complexity in log.
4. **Procedure/Algorithm: Pseudocode:**

```
if(y==0):  
    return 1;  
temp = power(x,y/2); if(y%2==0):  
    return temp*temp; else  
    return x*temp*temp;
```

Various possible cases

- 1)The power value can be 0
- 2)The power value can be negative.
- 3)The power value can be even.
- 4)The power value can be odd.

5. Code:

```
2
3 class PowerFunction {
4
5     static double power(double x, int y) { 5 usages
6         if (y == 0) {
7             return 1;
8         }
9
10        double temp = power(x, y / 2);
11
12        if (y % 2 == 0) {
13            return temp * temp;
14        } else {
15            if (y > 0) {
16                return x * temp * temp;
17            } else {
18                return (temp * temp) / x;
19            }
20        }
21    }
22
23    public static void main(String[] args) {
24        System.out.println("2^10 = " + power(2, 10));
25        System.out.println("2^-3 = " + power(2, -3));
26        System.out.println("5^0 = " + power(5, 0));
27        System.out.println("3^7 = " + power(3, 7));
28    }
29 }
```



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6. Output:

```
↓  
2^10 = 1024.0  
⇌  
2^-3 = 0.125  
⇌  
5^0 = 1.0  
⇌  
3^7 = 2187.0  
🖨  
🗑  
Process finished with exit code 0
```